UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 005 · Particular integral Type 3 (X = x^m) · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S005·8	Unit 2 deck 005 ("Particular integral of Type 3"), slide 8 (rendered .pptx)
T005 00:06:48	Unit 2 AV-summary transcript "Particular integral of standard functions – Type 3" (12.6 min), segment start
T006a 00:06:05	Unit 2 AV-summary transcript "Particular integral – Type 3 and Type 4" (bucket 006, 15.5 min), its Type 3 part 00:00:00–00:09:55; the class 006 pack cites the whole file as T006b
CW U2 P15	Unit 2 class-work problem 15
ESA Jul-23 3b	Past paper: UE22MA141A Jul 2023, question 3.b
ESA Jan-24 s2 2d	Past paper: UE23MA141A Jan 2024 set 2, question 2.d

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1 The Type 3 rule: $X = x^m$

Definitions

Type 3 right-hand side. $X = x^m$, m a positive integer; by superposition, any polynomial in x.

S005·3 · S005·4 · S005·10 · T005 00:00:18

Lowest-degree term. The term of $f(D)$ with the smallest power of D: the constant $a_0 = a_0 D^0$ when there is one ("minus 1 can be written as minus 1 into D to the power 0"), otherwise D or D^k.

S005·4 · T005 00:06:01

$\varphi(D)$**.** What is left of $f(D)$ after the lowest-degree term is taken out: $f(D) = a_0[1 \pm \varphi(D)]$. It contains only positive powers of D. S005·4 · T005 00:01:08

Statements in the class material

> *"Type (iii); When $X = x^n$, 'n' being a positive integer. Here PI $= \frac{1}{f(D)}(x^n)$. Take the lowest degree term common from $f(D)$ to get an expression in the form $(1 \pm \varphi(D))$ in the denominator. Take this factor to the numerator to get $(1 \pm \varphi(D))^{-1}$. Expand $(1 \pm \varphi(D))^{-1}$ using Binomial theorem upto nth power as $(n+1)^{\text{th}}$ derivative of x^n is zero. Operate on the numerator term by term by taking $D = \frac{d}{dx}$."* S005·4

> *"If $X = x^m$, then the particular integral is $y_p = \text{P.I} = \frac{1}{f(D)}x^m = [f(D)]^{-1}x^m$. Expand $[f(D)]^{-1}x^m$ in ascending powers of D using binomial theorem and operate on x^m term by term. Since $(m+1)$th term and higher derivatives are zero, it is not necessary to consider terms beyond D^m."* S005·10

Key points

1. Recognise Type 3: the right-hand side is an algebraic function, a polynomial in x ("$5x - 2$", "$x^2 + 7x + 9$"). ESA ★★★ T005 00:04:22 · S005·3
2. Take out the **lowest**-degree term of $f(D)$, with its sign and coefficient: $D^2 - 1 = -(1 - D^2)$, $2D^2 + 2D + 3 = 3\left[1 + \frac{2D^2+2D}{3}\right]$. ESA ★★★ S005·4 · S005·7 · S005·13 · T005 00:06:01 · T006a 00:08:22
3. Move $[1 \pm \varphi(D)]$ to the numerator as $[1 \pm \varphi(D)]^{-1}$; the constant $\frac{1}{a_0}$ stays in front. ESA ★★★ S005·4 · T005 00:01:21 · T005 00:09:27
4. Expand only up to D^m, where m is the degree of the polynomial (not the order of the equation): "the $(m+1)$th and higher derivatives are zero". ESA ★★★ S005·4 · S005·10 · T005 00:01:51 · T006a 00:05:30
5. Operate term by term: D differentiates once, D^2 twice, D^3 three times. ESA ★★★ S005·4 · T005 00:02:17 · T006a 00:07:02
6. If the question says "find the PI", the CF is not needed (slide 005·8); "solve" or "general solution" needs $y = y_c + y_p$. ESA ★★ S005·8 · T005 00:08:25 · S005·7
7. When $f(0) \neq 0$ the PI is a polynomial of the same degree as X; check the leading coefficient: it is the leading coefficient of X divided by $f(0)$. ESA ★★ ADDED FOR CLARITY S005·9 · S005·12

Why?

Why take out the lowest-degree term, not the highest?
Then $\varphi(D)$ has only positive powers of D, and the series is in ascending powers: each term is a derivative, and all but finitely many vanish on x^m. Taking out the highest power would give a series in $\frac{1}{D}$ (repeated integrals) that never ends.

ADDED FOR CLARITY S005·4 · T005 00:01:08

Why may the infinite series be cut at D^m?
$D^{m+1}x^m = 0$, and so is every higher derivative: "D^2, D^4, D^6 of $5x - 2$ is 0". The terms dropped contribute nothing, so the truncated operator gives the exact PI, not an approximation.

S005·4 · T005 00:07:49 · T006a 00:01:29

Why does a power series in D give a valid PI at all?
On polynomials of degree m, $f(D)$ acts like a polynomial in D with $D^{m+1} = 0$, so the truncated series is an exact inverse: $f(D)\,[\text{series}]\,x^m = x^m$. Substituting the PI back always returns X (checked with sympy for every example here).

ADDED FOR CLARITY S005·10

Exam traps

Exam trap. Slide 005·13: the roots of $2m^2 + 2m + 3 = 0$ are $\frac{-1\pm\sqrt{5}i}{2}$, not $\frac{-2\pm\sqrt{5}i}{2}$; slides 005·13–14 also drop c_2 in the CF. ADDED FOR CLARITY S005·13 · S005·14 · T006a 00:07:45

Exam trap. Do not forget the factor $\frac{1}{a_0}$ and its sign: taking -1 out of $D^2 - 1$ puts a minus in front of the whole PI. S005·7 · T005 00:06:17

Exam trap. Stop at D^m where m is the degree of X, not the order of $f(D)$: for x^5 and $D^3 - 1$ you need $1 + D^3$ (up to D^5), not just 1. ADDED FOR CLARITY CW U2 P16 · S005·10

Exam trap. T006a 00:02:23 says the nth derivative of x^m is zero "when n is m"; it is zero for $n > m$ ($D^m x^m = m!$). ADDED FOR CLARITY T006a 00:02:23 · T006a 00:01:29

2 The binomial expansions and where to stop

Definitions

Ascending powers of D. A series $1 + c_1D + c_2D^2 + \ldots$; applied to x^m it ends after D^m. S005·5 · S005·10 · T005 00:01:39

Truncation. Dropping every term of the expansion whose power of D exceeds m, since $D^{m+1}x^m = 0$. S005·4 · S005·10 · T006a 00:01:15

Statements in the class material

> *"The following expressions will be useful to expand $(1 \pm \varphi(D))^{-1}$ in ascending powers of D."* S005·5

Key points

1. Remember two series: $(1+z)^{-1} = 1 - z + z^2 - \ldots$ and $(1+z)^{-2} = 1 - 2z + 3z^2 - \ldots$; get the other two by $z \to -z$ (all signs plus). ESA ★★★ S005·5 · T005 00:03:12 · T005 00:04:04
2. Expand the square or cube of a two-term $\varphi(D)$ by $(a+b)^2$, $(a+b)^3$, but write down only the products with D-power $\leq m$. ESA ★★★ S005·9 · S005·12 · T005 00:10:49 · T006a 00:06:05
3. Collect the operator into $1 + c_1D + c_2D^2 + \cdots + c_mD^m$ before applying it: fewer derivatives, fewer slips. ESA ★★ (added for clarity) S005·9 · T006a 00:06:43
4. For $f(D) = (D-a)^2$ or $(D^2-1)^2$ use the $(1 \pm z)^{-2}$ series directly, instead of squaring the first-power series. ESA ★★ S005·5 · CW U2 P21 · CW U2 P22

Exam traps

Exam trap. Slide 005·7 writes $-(1-D^2)^{-1} = -(1 - D^2 + D^4 - \ldots)$; it is $-(1 + D^2 + D^4 + \ldots)$ (the lecturer corrects it, T005 00:07:07). The answer $2 - 5x$ survives only because $D^2(5x - 2) = 0$. (added for clarity) S005·7 · T005 00:07:07

Exam trap. In $(D^2 + 3D)^2$ the term $6D^3$ is kept when $m = 3$ but dropped when $m = 2$; decide product by product. T006a 00:06:22 · T005 00:11:24

3 When D is a factor of $f(D)$

Definitions

$\frac{1}{D}$. The inverse of D: integration with respect to x, without a constant of integration (class 003).
ADDED FOR CLARITY S005·4

Key points

1. If $f(D)$ has no constant term, its lowest-degree term is D^k: take it out, expand the rest, operate, then integrate k times. The PI has degree $m + k$. ESA ★ ADDED FOR CLARITY S005·4
2. When integrating ($\frac{1}{D}$) add no constant: constants belong to the CF, which already has c_1 (the root $m = 0$). ESA ★ ADDED FOR CLARITY S005·4

Why?

Why does a factor D raise the degree of the PI?
If $f(D) = D\psi(D)$, then $f(D)$ applied to a polynomial of degree m gives degree $m - 1$ at most, so to reproduce x^m the PI must have degree $m + 1$. The constant part of the PI is then a CF term ($m = 0$ is an AE root) and is dropped.
ADDED FOR CLARITY S005·4

Exam traps

Exam trap. With a factor D, do not add constants when integrating, and do not stop the expansion early: integrate after operating. ADDED FOR CLARITY S005·4

4 Mixed right-hand sides: Types 1, 2 and 3 together

Definitions

Mixed right-hand side. X is a sum of terms of different types, e.g. $e^{4x} + \sin x + x^2$. ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · ESA Jan-24 s4 2c

Key points

1. Split X into its terms first, label PI_1, PI_2, PI_3, and add at the end (E4.1). ESA ★★★ ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · ESA Jan-24 s4 2c
2. Find the CF first: its roots tell you which terms fail (a a root: e^{ax} fails; $\pm ai$ roots: $\sin ax$, $\cos ax$ fail; 0 a root: D is a factor for the polynomial). ESA ★★★ ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · CW U2 P21
3. $a^{-x} = e^{-x \log a}$, so 2^{-x} is Type 1 with $D \to -\log 2$: $(D^2 + 4) \to (\log 2)^2 + 4$. ESA ★★★ ESA Jan-24 s1 2d · ESA Jan-24 s4 2c
4. $\sin^2 \frac{x}{2} = \frac{1 - \cos x}{2}$ gives a constant (E3.2) and a Type 2 term; the polynomial x is a separate Type 3 term. ESA ★★★ ESA Jan-24 s4 2c · CW U2 P22

Exam traps

Exam trap. Never put x^2 into a Type 1 or Type 2 formula, and never expand in powers of D for e^{ax}: each term keeps its own rule. ADDED FOR CLARITY ESA Jul-23 3b

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences; 7 distinct / 9 occurrences are PI by operator rules). A Type 3 term sits in 4 distinct questions (4 occurrences), every one of them a mixed right-hand side: Jul-23 3b $(D^2-5D+6)y = e^{4x}+\sin x + x^2$ (7 marks), Jan-24 set 1 2d $(D^2+4)y = x^2 + \cos 2x + 2^{-x}$ (5), set 2 2d $(D^3-6D^2+11D-6)y = e^{2x}+x$ (5) and set 4 2c $(D-1)^2(D+1)^2y = \sin^2\frac{x}{2} + 2^{-x} + x$ (6, = CW P22). All four are solved in full here, with CW P15, P16 and P21. Stars: the Type 3 rule, the expansions and superposition ★★★ (4 questions); the case $f(D)$ with a factor D ★ (never asked); the second-order expansion $(1\pm z)^{-2}$ ★★ (CW P21 and P22 need it, no paper asks it on its own).

4 · Mixed right-hand sides: Types 1, 2 and 3 together

ESA Jul 2023 · 7 marks · UE22MA141A Q3.b · solved in W4.1

Solve $(D^2-5D+6)y = e^{4x}+\sin x + x^2$.

ESA Jan 2024 · 5 marks · UE23MA141A set 1, Q2.d · solved in W4.2

Solve $(D^2+4)y = x^2 + \cos 2x + 2^{-x}$.

ESA Jan 2024 · 5 marks · UE23MA141A set 2, Q2.d · solved in W4.3

Solve $(D^3-6D^2+11D-6)y = e^{2x}+x$.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q2.c (= CW P22) · solved in W4.4

Solve $[(D-1)^2(D+1)^2]y = \sin^2\frac{x}{2} + 2^{-x} + x$.